Semirings for database provenance

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What is a database?

Transactions

Transaction ID	Customer ID	Product ID	Purchase date	Quantity
43666	24221	389	17-02-2026	2
43666	24221	879	17-02-2026	3
43952	24222	975	17-02-2026	1
54673	24223	389	28-02-2026	2
58930	24224	975	09-03-2026	3
58930	24224	879	09-03-2026	5
59927	24223	789	10-03-2026	4

Products

Product ID	Product name	Price per kg
389	Avocado	5
789	Banana	2
879	Mango	6
975	Watermelon	3

Customers

Customer ID	First name	Last name
24221	Aditya	Prakash
24222	Graham	Leigh
24223	Ivan	Di Liberti
24224	Luigi	Brunetta

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- A *database* D as a finite set of relations.

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The *relational algebra* consists of the following operations:

- Selection, $\sigma_\theta(R)$: Keep tuples satisfying condition θ
- Projection, $\pi_L(R)$: Keep only attributes L
- Natural join, $R \bowtie S$: Merge tuples matching on common attributes
- Renaming, $\rho_\beta(R)$: Rename attributes according to β
- Union, $R \cup S$: Combine unique tuples
- Difference, $R - S$: Keep tuples present in R but not in S

Example

Find all the times a mango or banana was purchased

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Provenance is typically represented as a form of annotation information, usually by tagging data on the tuple-level of a relation.

What is a semiring?

Semiring framework

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The provenance for (a, a) from our query from the previous slide would be the polynomial $p^3 + 2pqr$.

Positive relational algebra

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- Difference

Aggregate queries

Books

title	genre	qty
book 1	adventure	4
book 2	fantasy	5
book 3	romance	6
book 4	adventure	3
book 5	fantasy	1
book 6	romance	2

SUM(qty)

GROUP BY genre

genre	total
adventure	7
fantasy	6
romance	8

Aggregation provenance

Difference in provenance

Approaches to difference

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- Quotient constructions

Open challenges

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- Unified framework for recursion, aggregates, and difference
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- Limits of semiring-based provenance